

closed surfaces obtained from M_1 and M_2 by filling in their boundary circle with a disk have smaller genus than M . Each of M_1 and M_2 is the mapping cylinder of a map from S^1 to a finite graph. Namely, view M_i as obtained from a closed surface by deleting an open disk in the interior of the 2-cell in the standard CW structure described in Chapter 0, so that M_i becomes the mapping cylinder of the attaching map of the 2-cell. This attaching map is not nullhomotopic, so it induces an injection on π_1 since free groups are torsionfree. Thus we have realized the original surface M as $K\Gamma$ for Γ a graph of groups of the form $F_1 \longleftarrow \mathbb{Z} \longrightarrow F_2$ with F_1 and F_2 free and the two maps injective. The theorem then says that M is a $K(G, 1)$.

A similar argument works for closed nonorientable surfaces other than $\mathbb{RP}^2$. For example, the Klein bottle is obtained from two Möbius bands by identifying their boundary circles, and a Möbius band is the mapping cylinder of the 2-sheeted covering space $S^1 \rightarrow S^1$.

Proof of 1B.11: We shall construct a covering space $\tilde{K} \rightarrow K\Gamma$ by gluing together copies of the universal covering spaces of the various mapping cylinders in $K\Gamma$ in such a way that $\tilde{K}$ will be contractible. Hence $\tilde{K}$ will be the universal cover of $K\Gamma$, which will therefore be a $K(G, 1)$.

First a preliminary observation: Given a universal covering space $p: \tilde{X} \rightarrow X$ and a connected, locally path-connected subspace $A \subset X$ such that the inclusion $A \hookrightarrow X$ induces an injection on π_1 , then each component $\tilde{A}$ of $p^{-1}(A)$ is a universal cover of A . To see this, note that $p: \tilde{A} \rightarrow A$ is a covering space, so the induced map $\pi_1(\tilde{A}) \rightarrow \pi_1(A)$ is injective, and this map factors through $\pi_1(\tilde{X}) = 0$, hence $\pi_1(\tilde{A}) = 0$. For example, if X is the torus $S^1 \times S^1$ and A is the circle $S^1 \times \{x_0\}$, then $p^{-1}(A)$ consists of infinitely many parallel lines in $\mathbb{R}^2$, each of which is a universal cover of A .

For a map $f: A \rightarrow B$ between connected CW complexes, let $p: \tilde{M}_f \rightarrow M_f$ be the universal cover of the mapping cylinder M_f . Then $\tilde{M}_f$ is itself the mapping cylinder of a map $\tilde{f}: p^{-1}(A) \rightarrow p^{-1}(B)$ since the line segments in the mapping cylinder structure on M_f lift to line segments in $\tilde{M}_f$ defining a mapping cylinder structure. Since $\tilde{M}_f$ is a mapping cylinder, it deformation retracts onto $p^{-1}(B)$, so $p^{-1}(B)$ is also simply-connected, hence is the universal cover of B . If f induces an injection on π_1 , then the remarks in the preceding paragraph apply, and the components of $p^{-1}(A)$ are universal covers of A . If we assume further that A and B are $K(G, 1)$'s, then $\tilde{M}_f$ and the components of $p^{-1}(A)$ are contractible, and we claim that $\tilde{M}_f$ deformation retracts onto each component $\tilde{A}$ of A . Namely, the inclusion $\tilde{A} \hookrightarrow \tilde{M}_f$ is a homotopy equivalence since both spaces are contractible, and then Corollary 0.20 implies that $\tilde{M}_f$ deformation retracts onto $\tilde{A}$ since the pair $(\tilde{M}_f, \tilde{A})$ satisfies the homotopy extension property, as shown in Example 0.15.

Now we can describe the construction of the covering space $\tilde{K}$ of $K\Gamma$. It will be the union of an increasing sequence of spaces $\tilde{K}_1 \subset \tilde{K}_2 \subset \dots$. For the first stage, let $\tilde{K}_1$ be the universal cover of one of the mapping cylinders M_f of $K\Gamma$. By the

preceding remarks, this contains various disjoint copies of universal covers of the two $K(G_v, 1)$'s at the ends of M_f . We build $\tilde{K}_2$ from $\tilde{K}_1$ by attaching to each of these universal covers of $K(G_v, 1)$'s a copy of the universal cover of each mapping cylinder M_g of $K\Gamma$ meeting M_f at the end of M_f in question. Now repeat the process to construct $\tilde{K}_3$ by attaching universal covers of mapping cylinders at all the universal covers of $K(G_v, 1)$'s created in the previous step. In the same way, we construct $\tilde{K}_{n+1}$ from $\tilde{K}_n$ for all n , and then we set $\tilde{K} = \bigcup_n \tilde{K}_n$.

Note that $\tilde{K}_{n+1}$ deformation retracts onto $\tilde{K}_n$ since it is formed by attaching pieces to $\tilde{K}_n$ that deformation retract onto the subspaces along which they attach, by our earlier remarks. It follows that $\tilde{K}$ is contractible since we can deformation retract $\tilde{K}_{n+1}$ onto $\tilde{K}_n$ during the time interval $[1/2^{n+1}, 1/2^n]$, and then finish with a contraction of $\tilde{K}_1$ to a point during the time interval $[1/2, 1]$.

The natural projection $\tilde{K} \rightarrow K\Gamma$ is clearly a covering space, so this finishes the proof that $K\Gamma$ is a $K(G, 1)$.

The remaining statement that each inclusion $K(G_v, 1) \hookrightarrow K\Gamma$ induces an injection on π_1 can easily be deduced from the preceding constructions. For suppose a loop $\gamma: S^1 \rightarrow K(G_v, 1)$ is nullhomotopic in $K\Gamma$. By the lifting criterion for covering spaces, there is a lift $\tilde{\gamma}: S^1 \rightarrow \tilde{K}$. This has image contained in one of the copies of the universal cover of $K(G_v, 1)$, so $\tilde{\gamma}$ is nullhomotopic in this universal cover, and hence γ is nullhomotopic in $K(G_v, 1)$. $\square$

The various mapping cylinders that make up the universal cover of $K\Gamma$ are arranged in a treelike pattern. The tree in question, call it $T\Gamma$, has one vertex for each copy of a universal cover of a $K(G_v, 1)$ in $\tilde{K}$, and two vertices are joined by an edge whenever the two universal covers of $K(G_v, 1)$'s corresponding to these vertices are connected by a line segment lifting a line segment in the mapping cylinder structure of a mapping cylinder of $K\Gamma$. The inductive construction of $\tilde{K}$ is reflected in an inductive construction of $T\Gamma$ as a union of an increasing sequence of subtrees $T_1 \subset T_2 \subset \dots$. Corresponding to $\tilde{K}_1$ is a subtree $T_1 \subset T\Gamma$ consisting of a central vertex with a number of edges radiating out from it, an 'asterisk' with possibly an infinite number of edges. When we enlarge $\tilde{K}_1$ to $\tilde{K}_2$, T_1 is correspondingly enlarged to a tree T_2 by attaching a similar asterisk at the end of each outer vertex of T_1 , and each subsequent enlargement is handled in the same way. The action of $\pi_1(K\Gamma)$ on $\tilde{K}$ as deck transformations induces an action on $T\Gamma$, permuting its vertices and edges, and the orbit space of $T\Gamma$ under this action is just the original graph Γ . The action on $T\Gamma$ will not generally be a free action since the elements of a subgroup $G_v \subset \pi_1(K\Gamma)$ fix the vertex of $T\Gamma$ corresponding to one of the universal covers of $K(G_v, 1)$.

There is in fact an exact correspondence between graphs of groups and groups acting on trees. See [Scott & Wall 1979] for an exposition of this rather nice theory. From the viewpoint of groups acting on trees, the definition of a graph of groups is

usually taken to be slightly more restrictive than the one we have given here, namely, one considers only oriented graphs obtained from an unoriented graph by subdividing each edge by adding a vertex at its midpoint, then orienting the two resulting edges outward, away from the new vertex.

Exercises

1. Suppose a group G acts simplicially on a Δ -complex X , where ‘simplicially’ means that each element of G takes each simplex of X onto another simplex by a linear homeomorphism. If the action is free, show it is a covering space action.
2. Let X be a connected CW complex and G a group such that every homomorphism $\pi_1(X) \rightarrow G$ is trivial. Show that every map $X \rightarrow K(G, 1)$ is nullhomotopic.
3. Show that every graph product of trivial groups is free.
4. Use van Kampen’s theorem to compute $A *_C$ as a quotient of $A * \mathbb{Z}$, as stated in the text.
5. Consider the graph of groups Γ having one vertex, $\mathbb{Z}$, and one edge, the map $\mathbb{Z} \rightarrow \mathbb{Z}$ that is multiplication by 2, realized by the 2-sheeted covering space $S^1 \rightarrow S^1$. Show that $\pi_1(K\Gamma)$ has presentation $\langle a, b \mid bab^{-1}a^{-2} \rangle$ and describe the universal cover of $K\Gamma$ explicitly as a product $T \times \mathbb{R}$ with T a tree. [The group $\pi_1(K\Gamma)$ is the first in a family of groups called Baumslag-Solitar groups, having presentations of the form $\langle a, b \mid ba^mb^{-1}a^{-n} \rangle$. These are HNN extensions $\mathbb{Z} *_Z$.]
6. Show that for a graph of groups all of whose edge homomorphisms are injective maps $\mathbb{Z} \rightarrow \mathbb{Z}$, we can choose $K\Gamma$ to have universal cover a product $T \times \mathbb{R}$ with T a tree. Work out in detail the case that the graph of groups is the infinite sequence $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{3} \mathbb{Z} \xrightarrow{4} \mathbb{Z} \rightarrow \cdots$ where the map $\mathbb{Z} \xrightarrow{n} \mathbb{Z}$ is multiplication by n . Show that $\pi_1(K\Gamma)$ is isomorphic to $\mathbb{Q}$ in this case. How would one modify this example to get $\pi_1(K\Gamma)$ isomorphic to the subgroup of $\mathbb{Q}$ consisting of rational numbers with denominator a power of 2?
7. Show that every graph product of groups can be realized by a graph whose vertices are partitioned into two subsets, with every oriented edge going from a vertex in the first subset to a vertex in the second subset.
8. Show that a finite graph product of finitely generated groups is finitely generated, and similarly for finitely presented groups.
9. Show that a finite graph product of finite groups has a free subgroup of finite index, by constructing a finite-sheeted covering space of $K\Gamma$ from universal covers of the mapping cylinders of $K\Gamma$. [The converse is also true for finitely generated groups; see [Scott & Wall 1979] for more on this.]